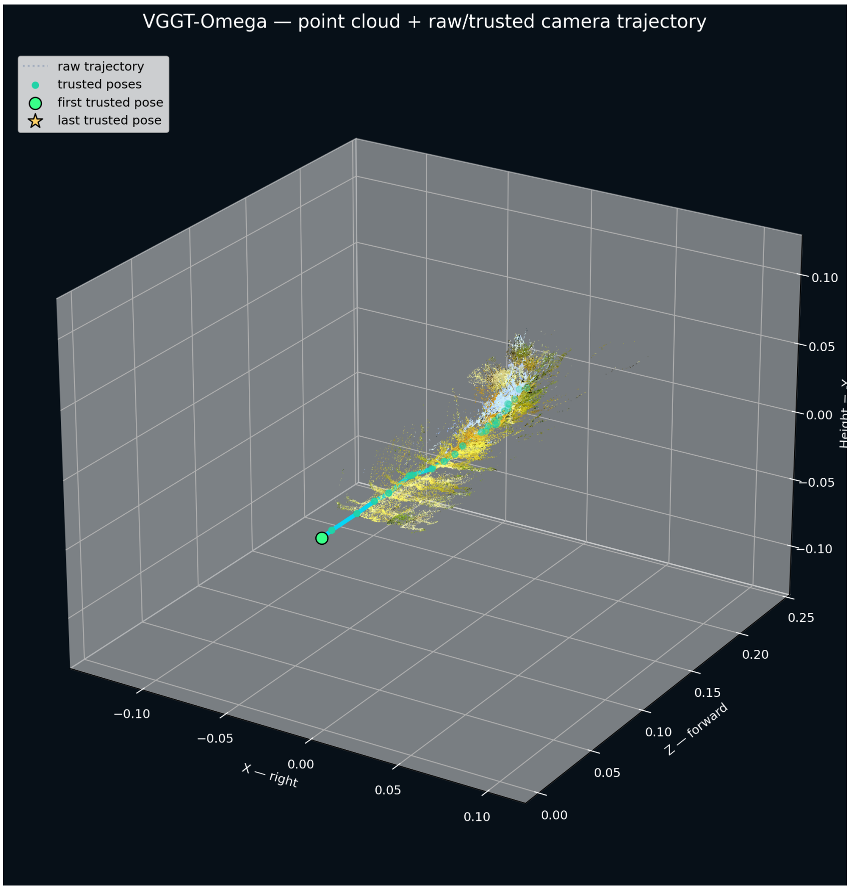
PROJECT GUIDE

FPV Reconstruction

From one video link to an auditable 3D scene and relative camera path



STATUS: Stable relative-reconstruction baseline, ready for demonstration and review.

- One direct video URL starts the complete workflow.
- Four representative videos completed; 119 of 120 model poses were trusted.
- Every run produced the main 3D, image, trajectory, and quality outputs.

Important: XYZ is relative reconstruction space. It is not metres, physical speed, latitude, longitude, or absolute altitude.

THE WORKFLOW

What the notebook does

The notebook turns an FPV flight into a checked reconstruction while keeping the technical work out of the main user view.

1 INPUT

Paste one direct video URL.

2 INSPECT

Read timing, resolution, frame rate, and decode health.

3 CLEAN

Remove freezes and unusable intervals without inventing frames.

4 SELECT

Choose compact views with scene coverage and overlap.

5 RECONSTRUCT

Estimate depth, scene points, intrinsics, and poses.

6 CHECK AND EXPORT

Create GLB, previews, XYZ tables, warnings, and ZIP.

USE: Paste the link, select a T4 GPU or better, choose Runtime -> Run all, and review the important outputs together in Results.

Why the result is auditable

- Selected frames, raw coordinates, trusted coordinates, and warnings remain available.
- Weak evidence is marked instead of silently hidden.
- A convincing point cloud must agree with the source frames and trajectory checks.

One canonical baseline

The validated logic is kept in one notebook. Difficult clips are handled through evidence and warnings instead of separate hidden versions.

VIDEO PREPARATION

Cleaning without fabricating motion

FPV recordings may contain frozen images, repeated frames, cuts, compression damage, or unreadable frames. Treating these as real motion can mislead reconstruction.


Early retained view


Mid-flight retained view


Late retained view

Freeze detection in plain language

The notebook compares consecutive frames. When many frames remain nearly identical one after another, the interval is treated as a freeze rather than genuine flight. The frozen interval is removed and recorded.

What happens at a removed interval

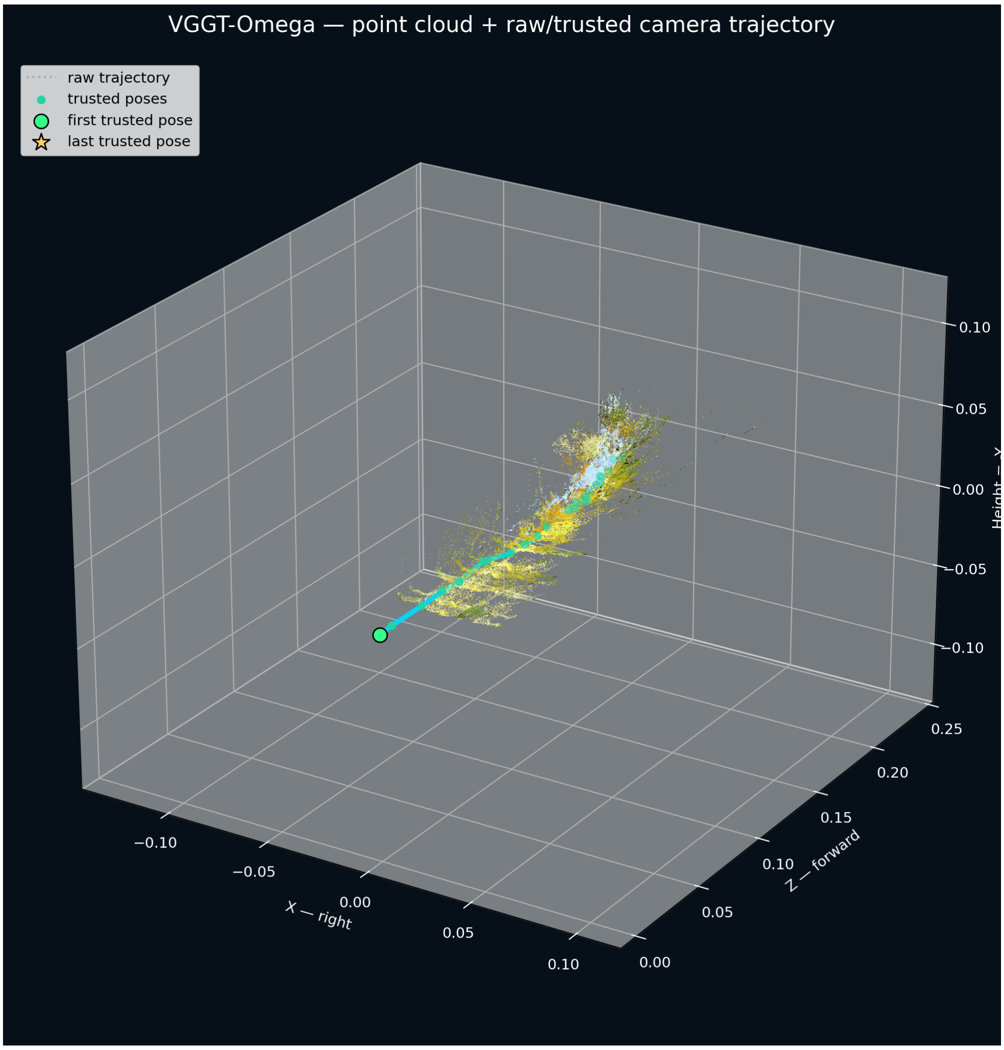
- Usable frames before and after the cut remain available.
- The notebook never fills missing positions with a duplicate frame.
- The trajectory does not interpolate across unobserved time.
- Timing and segment evidence remain in the audit files.

> **RESULT:** The cleaned video is more useful while the removed information remains visible in the evidence.

3D RECONSTRUCTION

How VGGT-Omega contributes

VGGT-Omega receives selected overlapping views and estimates camera poses, camera intrinsics, depth, and 3D scene points. The notebook converts these estimates into the GLB scene and relative XYZ path.



MODEL ESTIMATES

Camera locations, depth, intrinsics, and colored scene points in one relative coordinate system.

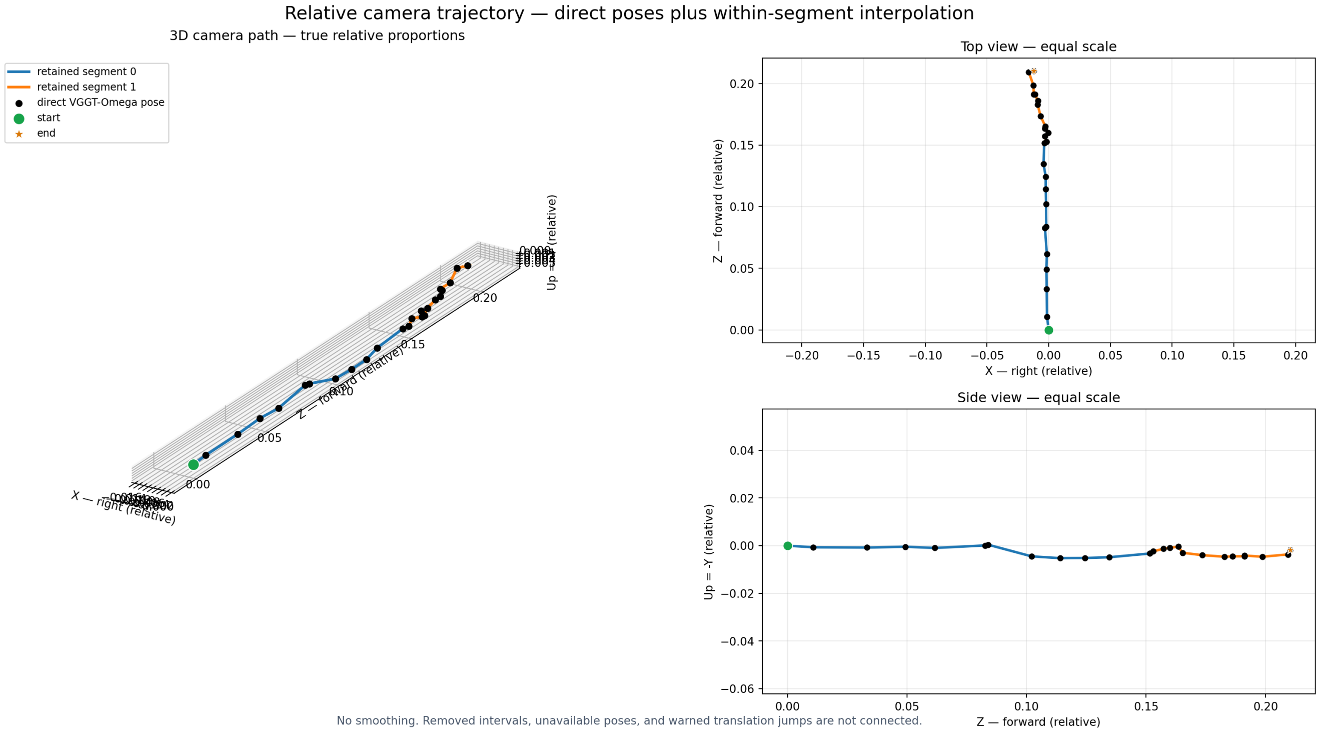
NOTEBOOK CHECKS

Pose trust, continuity, geometry components, unusual motion, GPU quality, and output completeness.

The first camera is the origin. This defines the relative reference system; it does not mean the start location is geographically known.

CAMERA PATH

A true-scale trajectory, without invented connections



The result contains an orthographic true-proportion 3D view and equal-scale top and side views. Small sideways movement is no longer stretched until it looks comparable to forward motion.

- Black markers are direct VGGT-Omega poses; colors separate retained video segments.
- Direct poses are not smoothed.
- Removed intervals, unavailable poses, and warned translation jumps are not connected.
- Every raw pose remains in CSV for inspection.

> **WHY THIS MATTERS:** The chart now communicates the measured geometry honestly without altering VGGT output.

EVIDENCE AND DELIVERY

What is ready now

Video	Trusted	Geometry	Result	Main observation
Adaissah	25 / 25	1 component	**PASS***	Coherent mostly-forward path.
Namer / Khiam	27 / 27	1 component	**PASS***	Warnings match visible hard motion.
Kfar Tebnit	40 / 40	1 component	**PASS***	Known jump is warned and disconnected.
Majdal Zoun	27 / 28	1 component	**PASS***	One weak final pose is masked.

* PASS with visible warnings retained for review.

Main outputs

GLB scene | reconstruction PNG | selected-frame overview | true-scale trajectory PNG | raw and trusted XYZ CSV | per-frame CSV | audits | summary JSON | ZIP

Open the project

GitHub repository
Shared Google Colab notebook
Adaissah example outputs

CURRENT BOUNDARY: The baseline establishes relative reconstruction. Metric scale and geographic alignment are separate next-stage decisions after review.